

Assignment 2

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Cloud Networking
Development*

Public Portfolio Edition

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Introduction

Containerization has become an important part of modern software development because it allows applications to run the same way on different computers. Docker is one of the most popular tools used for this purpose. It packages an application together with everything it needs to run. This removes many installation and compatibility problems and makes it easier to share software with other users.

This project focuses on creating a containerized console application that determines whether a student qualifies to write an examination. The application calculates a weighted average using the student's Test 1, Test 2, Assignment 1 and Project marks. The completed application was packaged into a Docker image and uploaded to Docker Hub. This document explains what the application does and provides the steps needed to download and run it using Docker.

Student Qualifier Application

1. Project Overview

The Student Qualifier Application is a console program that determines whether a student qualifies to write an examination.

The user enters four assessment marks:

- Test 1
- Test 2
- Assignment 1
- Project

The program calculates the weighted average using the assessment weights shown below.

Assessment	Weight
Test 1	30%
Test 2	50%
Assignment 1	10%
Project	10%

After calculating the final mark, the application displays whether the student qualifies to write the examination. A student qualifies if the final weighted average is **50% or higher**.

2. Technologies Used

The following technologies were used to develop the application.

- C#
- .NET Console Application
- Docker
- Docker Hub

3. Docker Hub Repository

The Docker image for this project has been uploaded to Docker Hub.

Repository

`student-qualifier`

Docker Hub Repository

Public source and Dockerfile are included in this repository. `student-qualifier`

4. Requirements

Before running the application, make sure the following requirements are met.

- Docker Desktop is installed.
- Docker Desktop is running.
- An internet connection is available to download the Docker image.

5. Download the Docker Image

Open a terminal or command prompt and run the following command.

```
docker pull student-qualifier:1.0
```

This command downloads version **1.0** of the Student Qualifier application from Docker Hub.

6. Run the Application

After the image has been downloaded, run the following command.

```
docker run -it student-qualifier:1.0
```

The `-it` option starts the container in interactive mode. This allows the user to enter the assessment marks through the console.

7. How the Application Works

The application performs the following steps.

1. Requests the Test 1 mark.
2. Requests the Test 2 mark.
3. Requests the Assignment 1 mark.
4. Requests the Project mark.
5. Calculates the weighted average.
6. Displays the final weighted average.
7. Displays whether the student qualifies to write the examination.

8. Example

User Input

```
Test 1: 70  
Test 2: 60  
Assignment 1: 80  
Project: 75
```

Weighted Average Calculation

$$(70 \times 30\%) + (60 \times 50\%) + (80 \times 10\%) + (75 \times 10\%)$$

= 66.5%

Output

Weighted Average: 66.5%

Status:

Student qualifies to write the examination.

9. Conclusion

The Student Qualifier Application provides a simple way to calculate whether a student qualifies to write an examination. Docker allows the application to run on any computer with Docker Desktop installed. Users do not need to install C# or any other software. They only need to download the Docker image and run it using the Docker commands provided in this document.

References

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